

# VELOCITY DRIVES GREATER POWER OBSERVED DURING BACK SQUAT USING CLUSTER SETS

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## ABSTRACT

Oliver, JM, Kreutzer, A, Jenke, SC, Phillips, MD, Mitchell, JB, and Jones, MT. Velocity drives greater power observed during back squat using cluster sets. *J Strength Cond Res* 30(1): 235–243, 2016—This investigation compared the kinetics and kinematics of cluster sets (CLU) and traditional sets (TRD) during back squat in trained (RT) and untrained (UT) men. Twenty-four participants (RT = 12, 25 ± 1 year, 179.1 ± 2.2 cm, 84.6 ± 2.1 kg; UT = 12, 25 ± 1 year, 180.1 ± 1.8 cm, 85.4 ± 3.8 kg) performed TRD (4 × 10, 120-second rest) and CLU (4 × (2 × 5) 30 seconds between clusters; 90 seconds between sets) with 70% one repetition maximum, randomly. Kinematics and kinetics were sampled through force plate and linear position transducers. Resistance-trained produced greater overall force, velocity, and power; however, similar patterns were observed in all variables when comparing conditions. Cluster sets produced significantly greater force in isolated repetitions in sets 1–3, while consistently producing greater force due to a required reduction in load during set 4 resulting in greater total volume load (CLU, 3302.4 ± 102.7 kg; TRD, 3274.8 ± 102.8 kg). Velocity loss was lessened in CLU resulting in significantly higher velocities in sets 2 through 4. Furthermore, higher velocities were produced by CLU during later repetitions of each set. Cluster sets produced greater power output for an increasing number of repetitions in each set (set 1, 5 repetitions; sets 2 and 3, 6 repetitions; set 4, 8 repetitions), and the difference between conditions increased over subsequent sets. Time under tension increased over each set and was greater in TRD. This study demonstrates greater power output is driven by greater velocity when back squatting during CLU; therefore, velocity may be a useful measure by which to assess power.

**KEY WORDS** biomechanics, exercise, kinesiology, performance, resistance

## INTRODUCTION

Traditionally, when performing resistance training, repetitions are performed in a continuous manner resulting in a decrease in velocity (25) and thus power output. In sports where the ability to generate power is a necessary aspect of performance, the resultant decrease in power output is counter to the principle of specificity. One method to counter the reduction in velocity (18,23) and power output (15,23,29) is the use of cluster sets (CLU) or clustering. Cluster sets incorporate short periods of rest between individual repetitions (interrepetition rest) or clusters of repetitions (intrasets rest). It has been hypothesized that 15–30 seconds of recovery between repetitions would allow for partial replenishment of phosphocreatine (PCr) stores (24), thus facilitating sufficient recovery to allow augmented quality of movement in later repetitions. This has been confirmed in studies that have demonstrated greater peak velocity in the clean pull at 90 and 120% of one repetition maximum (1RM) (18), greater total power output at 85% of 1RM in the bench press (29), and greater force, velocity, and power output during the power clean at 80% of 1RM (23) when using CLU as compared with TRD. Furthermore, acute studies have provided support to the theory of partial PCr repletion, showing greater ATP and PCr concentrations (15) and less reliance on anaerobic glycolysis (8,14) during CLU.

Findings on the effects of CLU on training adaptations are equivocal (9,20,26,28). A deciding factor regarding the effectiveness of CLU in the augmentation of power development seems to be exercise intensity. Studies in which intensities used were close to the optimal load (4) for power development (e.g., 80% 1RM power clean) in the respective exercise showed beneficial effects of CLU (5,11,20,26,32,36) on power output, while those working above the optimal load did not (9,28). However, a number of those studies also demonstrated CLU to produce smaller strength gains (20,36). Oliver et al. (32) recorded greater strength and power development in the back squat and bench press after 12 weeks of hypertrophic training with CLU when compared with TRD in a population of resistance-trained (RT) men (≥2 years). Furthermore, the authors reported no differences in lean mass gains, suggesting that CLU were as

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effective in increasing lean as TRD while augmenting power and strength development when using intensities corresponding to hypertrophy. This may have been a result of the intensities used (65–75% 1RM). However, those results were in conflict with Goto et al. (16) who reported smaller gains in lean mass and strength in non-RT recreationally active subjects after CLU using similar intensities (~75% 1RM). Moreover, Folland et al. (11) observed similar isometric strength gains after CLU and TRD in untrained and a nonsignificant tendency ( $p < 0.10$ ) toward greater high-velocity strength gains in CLU. The associated kinematics (i.e., force, velocity, and power) and kinetics (i.e., time under tension, concentric time under tension, and eccentric time under tension) were not examined in those previous studies in which differing adaptations were reported. Given that studies that have demonstrated a beneficial effect of CLU have incorporated intensities around those prescribed for hypertrophy (65–80% 1RM), it may be that differences in kinetics and kinematics may be contributing to divergent adaptations between those of varying training status.

It has been proposed that the kinetics and kinematics are the most important stimuli responsible for the resultant resistance training-induced adaptation (31). To the authors' knowledge, a direct comparison of the effects of CLU in RT and untrained (UT) individuals on the kinetics and kinematics has not been examined. Therefore, the purpose of this study was to examine the acute effects of CLU and TRD on the kinetics and kinematics during the back squat in young RT and UT men.

## METHODS

### Experimental Design

This study followed a repeated-measures, counterbalanced, and randomized design in which each participant served as his own control completing both experimental conditions. After determination of height, weight, and body composition, participants completed a familiarization session. At

least 48 hours after the familiarization, participants reported to the laboratory for determination of one 1RM. A minimum of 72 hours after 1RM determination, participants underwent the same experimental testing procedures under 2 conditions, randomly assigned: TRD and CLU. At least 7 days separated the experimental conditions.

### Subjects

Twenty-four participants ( $n = 24$ ) completed this study of which 12 were RT (RT = 12) having  $\geq 3$  years' experience performing the back squat exercise and 12 were untrained (UT = 12). Selection criteria included (a) men between the ages of 20 and 35 years of age (b) with no history of smoking and/or tobacco use (6 months), (c) not taking thyroid, androgenic, or other medications known to affect endocrine function, and (d) reportedly not consuming any ergogenic levels of nutritional supplements known affect muscle mass, insulin-like substances, or anabolic/catabolic prohormones or hormones within the previous 6 months leading up to the study. All procedures involving human participants were approved by the Institutional Review Board of Texas Christian University for the use of human participants in research. Written consent was obtained from all participants. Baseline demographics are presented in Table 1. Although both groups were similar in age, height, and body mass, UT participants had significantly higher body fatness as determined by dual x-ray absorptiometry (DXA) ( $p < 0.001$ ). As expected, 1RM parallel back squat ( $p < 0.001$ ) and 1RM parallel back squat to body weight ratio ( $p < 0.001$ ) was significantly greater in RT when compared with UT.

### Familiarization Session

Before familiarization, participants' height and body mass were determined to the nearest 0.1 cm and 0.1 kg, respectively, using a stadiometer (Seca, Chino, CA, USA) and self-calibrating digital scale (Seca) with participants in socks or bare feet. Participants then underwent body composition determination through DXA (GE Healthcare, Little Chalfont, United Kingdom) calibrated according to manufacturer's guidelines and performed by a trained technician. All participants participated in a familiarization session. After research personnel demonstrated proper form for the back squat exercise, participants were required to perform the exercise until demonstrating proficiency. Only the weight of the bar (20.4 kg) was used during familiarization. Those unable to perform using proper form were excluded from further testing.

**TABLE 1.** Baseline demographics.\*†

|                           | UT ( $n = 12$ ) | RT ( $n = 12$ )  |
|---------------------------|-----------------|------------------|
| Age (y)                   | 25.0 $\pm$ 0.9  | 24.7 $\pm$ 1.2   |
| Height (cm)               | 180.1 $\pm$ 1.8 | 179.1 $\pm$ 2.2  |
| Body mass (kg)            | 85.4 $\pm$ 3.8  | 84.6 $\pm$ 2.1   |
| Body fat (%)              | 28.3 $\pm$ 2.1  | 15.8 $\pm$ 1.3‡  |
| Lean mass (kg)            | 57.4 $\pm$ 1.6  | 67.7 $\pm$ 1.6‡  |
| Bone mineral content (kg) | 3.1 $\pm$ 0.12  | 3.6 $\pm$ 0.3‡   |
| Training age (y)          | 0.0 $\pm$ 0.0   | 6.1 $\pm$ 0.6‡   |
| 1RM back squat (kg)       | 90.0 $\pm$ 5.6  | 146.9 $\pm$ 4.9‡ |
| 1RM back squat: body mass | 1.1 $\pm$ 0.1   | 1.7 $\pm$ 0.1‡   |

\*RT = resistance-trained men; UT = untrained men.

†Values are means  $\pm$  SE.

‡Significant difference between groups.

### One Repetition Maximum Testing

At least 48 hours after familiarization, participants returned to the laboratory for determination of 1RM in the back squat exercise. They were instructed to refrain from any physical activity outside of daily living for the previous 48 hours. After a dynamic warm-up (8–10 minutes), participants performed 2 sets of 5 repetitions at 40–60% of their estimated 1RM with 2-minute rest between sets. After 3-minute rest, participants performed 1 to 2 sets of 2–3 repetitions at a load corresponding to 60–80% 1RM. Participants then began performing sets of 1 repetition of increasing weight for 1RM determination. Three to 5 minutes of rest was provided between each successive attempt. All 1RM determinations were made within 3 to 5 attempts. Participants were required to reach a depth of the squat at which the top of the thighs were parallel to the floor as determined by research personnel for an attempt to be considered successful. A verbal “up” command was provided. 1RM was defined as the point at which the participant could no longer increase the weight and complete a full repetition while maintaining proper form. For all 1RM testing, safety bars were put in place to prevent injury. This method of 1RM determination has been shown to have an intraclass coefficient of 0.99 and a corresponding Pearson product-moment coefficient of 0.001 (32). At the end of the final repetition, placement of both feet was measured and recorded. During a subsequent repetition using only the bar (20.4 kg), participants were asked to pause at the bottom of the repetition to mark parallel depth. Foot placement and parallel depth were used in all subsequent testing. All testing was performed on an Optima Smith Machine (LifeFitness, Schiller Park, IL, USA).

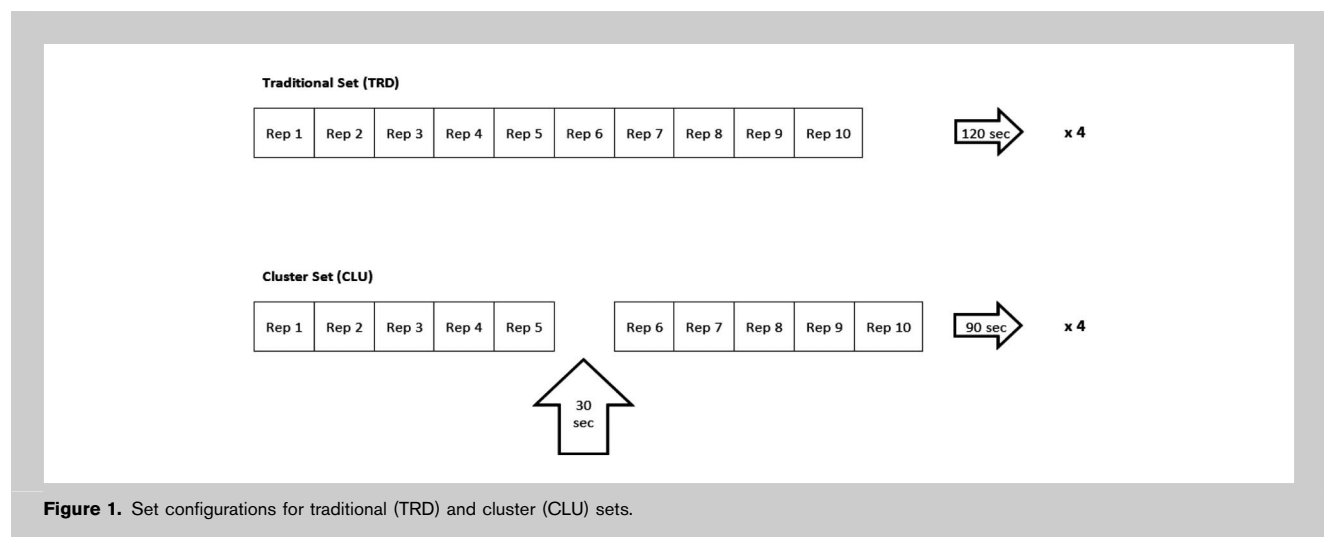
### Experimental Testing

Participants reported to the laboratory having refrained from any lower body training for the previous 72 hours and any activities outside of daily living for the previous 48 hours. After a dynamic warm-up (8–10 minutes), participants

performed 2 sets of 5 repetitions of the back squat exercise with a load equivalent to 40 and 60% of 1RM. After 2 minutes of rest, participants performed the back squat using either TRD or CLU with a load equivalent to 70% of 1RM. The set configurations (Figure 1) were as follows: (a) TRD consisted of 4 sets of 10 repetitions ( $4 \times 10$ ) with 120 seconds of rest between sets and (b) CLU consisted of 4 sets of 2 clusters of 5 repetitions ( $4 \times (2 \times 5)$ ) with 30 seconds of rest between clusters and 90 seconds of rest between sets. Participants were instructed to perform the concentric (upward) portion of each repetition “as explosively as possible.” Verbal encouragement was provided throughout the trial. If a participant paused for more than 2 seconds in the extended position or was unable to complete a repetition, resistance was lowered by 13.6 kg.

### Force Plate and Linear Position Transducers

Participants performed all exercise bouts on an AccuPower portable force platform (Advanced Mechanical Technology, Inc., Watertown, MA, USA) with the right side of the barbell attached to 2 linear position transducers (LPT) (Advanced Mechanical Technology, Inc.). The LPTs were mounted below and anterior and below and posterior to the participant, forming a triangle when attached to the barbell, thus allowing for measurement of horizontal and vertical bar displacement. The LPTs produced a voltage signal that represented the degree at which the LPTs were extended, allowing for the calculation of displacement-time data (3). From this displacement-time data, instantaneous velocity was calculated throughout the movement. Ground reaction force collected through force plate and displacement data were sampled at 1000 Hz by an analog-to-digital converter (Sewell Direct, Provo, UT, USA) and collected by a laptop computer using custom-built data acquisition and analysis software (Treadmetrix, Park City, UT, USA). Peak and average values for each individual repetition and for each set were determined for force and velocity. Average power



was calculated as the product of average force and velocity for each repetition over all sets. The reliability of the equipment and software was assessed through comparison of average power between 2 trials over 2 repetitions. The intra-class correlation coefficient for this comparison was 0.97 ( $p = 0.001$ ). Concentric (CON), eccentric (ECC), and total (TOT) time under tension (TUT) was calculated from the displacement-time data for each individual repetition.

### Statistical Analyses

All analyses were performed using Statistical Package for the Social Sciences (SPSS) version 22 (IBM Corp., Armonk, NY, USA). All data were normally distributed as determined by the Kolomogrov-Smirnov test of normality. A multifactorial analysis of variance with repeated measures was used to determine the statistical significance of our findings. The factors included training status (2 levels), condition (2 levels), and time (multiple levels depending on the variable in question). Bonferroni post hoc analysis was performed when a significant finding ( $p \leq 0.05$ ) or trend ( $p \leq 0.10$ ) was identified. Dependent measures were force, velocity, and power output as well as CON, ECC, and TOT TUT, load, and total volume load (TVL) calculated as the product of set  $\times$  repetitions  $\times$  load. Data are presented as mean  $\pm$  SE ( $s_x$ ), unless otherwise noted.

## RESULTS

Overall, similar patterns were observed in all kinematic variables for both RT and UT when comparing TRD with CLU as evidenced by only a main effect of training status with no significant interactions with training status observed.

### Kinematic Profile: Force, Velocity, and Power

A significant condition by set by repetition ( $p < 0.001$ ) interaction was observed when examining average force in which CLU resulted in significantly greater average force in isolated repetitions in set 1 (rep 1,  $p = 0.041$ ; rep 8,  $p = 0.053$ ) and set 3 (rep 7,  $p = 0.028$ ); a pattern was observed in set 2 (rep 7,  $p = 0.018$ ; rep 8,  $p = 0.014$ ; rep 9,  $p = 0.015$ ) and set 4 (rep 7,  $p = 0.003$ ; rep 8,  $p = 0.001$ ; rep 9,  $p < 0.001$ ; rep 10,  $p < 0.001$ ) in which CLU consistently produced greater average force in later repetitions during those sets. Average force collapsed across all sets in presented in Table 2. When collapsed across all sets, CLU produced greater average force in reps 1, 6 through 10.

Significant condition by set ( $p = 0.014$ ) and condition by repetition ( $p < 0.001$ ) interactions were observed when examining average velocity. When collapsed across all repetitions, velocity decreased from sets 1 to 4 in both conditions (Figure 2A), but the rate of decline was lessened in CLU resulting in higher velocities in sets 2 through 4. Figure 2B shows the velocity for each repetition collapsed across all sets. Cluster sets resulted in a significantly higher velocity for repetition (rep) 1 and all later repetitions (repetitions 6–10). Furthermore, a trend was noted in the 3-way interaction of condition by set by rep ( $p = 0.10$ ). Post hoc analysis on the

**TABLE 2.** Average force output (N) collapsed across sets.\*†

| Repetition | TRD           | CLU            |
|------------|---------------|----------------|
| 1          | 1660 $\pm$ 35 | 1665 $\pm$ 35‡ |
| 2          | 1665 $\pm$ 34 | 1668 $\pm$ 35§ |
| 3          | 1665 $\pm$ 35 | 1669 $\pm$ 35  |
| 4          | 1664 $\pm$ 35 | 1668 $\pm$ 35  |
| 5          | 1663 $\pm$ 35 | 1667 $\pm$ 35  |
| 6          | 1661 $\pm$ 35 | 1666 $\pm$ 35§ |
| 7          | 1654 $\pm$ 36 | 1667 $\pm$ 35‡ |
| 8          | 1650 $\pm$ 36 | 1666 $\pm$ 35‡ |
| 9          | 1643 $\pm$ 36 | 1664 $\pm$ 35‡ |
| 10         | 1641 $\pm$ 37 | 1663 $\pm$ 35‡ |

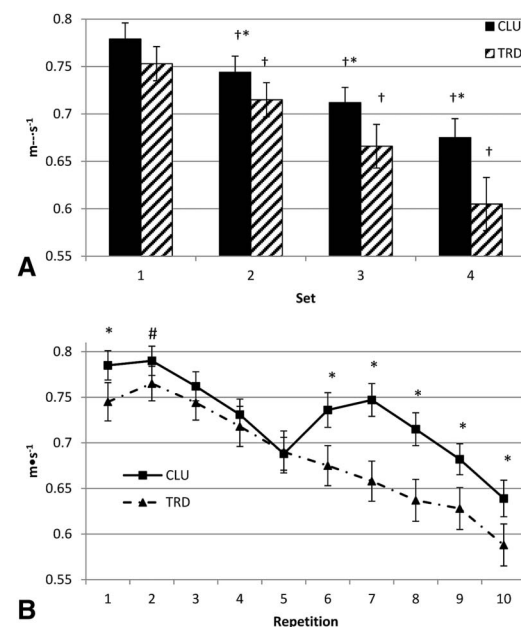
\*TRD = traditional sets; CLU = cluster sets.

†Values are means  $\pm$  SE.

‡Significant difference between conditions ( $p \leq 0.05$ ).

§Significant difference between conditions ( $p < 0.10$ ).

observed trend showed that the number of repetitions in which CLU produced greater velocity increased with each subsequent set (set 1: repetition 7, 10; set 2: repetition 7, 8, 9, 10; set 3: repetitions 1, 6, 7, 8, 9; set 4: repetitions 1, 2, 3, 6, 7, 8, 9).



**Figure 2.** Average velocity ( $\text{m} \cdot \text{s}^{-1}$ ) collapsed by training status across reps (A) and across sets (B). †Significantly different from set 1 in same condition ( $p \leq 0.05$ ); \*Significant difference between conditions ( $p \leq 0.05$ ); #Significant difference between conditions ( $p < 0.10$ ).

A condition by set by rep ( $p = 0.033$ ) interaction was observed for average power output. Cluster sets resulted in greater average power output for an increasing number of repetitions in each set. Participants produced greater average power when performing CLU in 5 repetitions (1,7–10) during set 1, 6 repetitions (1,6–10) during sets 2 and 3, and 8 repetitions (1–3,6–10) during set 4. Furthermore, the differ-

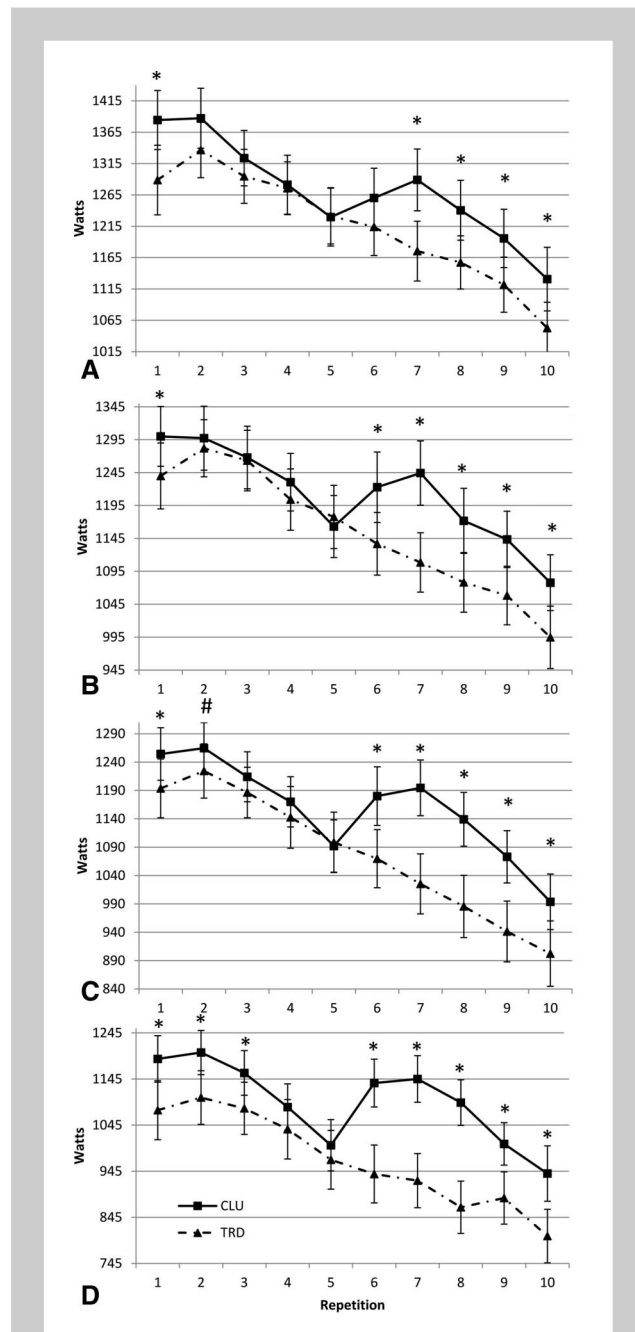
ence in average power output in repetitions 6 through 10 between CLU and TRD became increasingly larger from set to set (Figure 3A–D).

#### Kinetic Profile: Time Under Tension, Concentric Time Under Tension, and Eccentric Time Under Tension

A main effect of set ( $p < 0.001$ ) and a condition by repetition ( $p < 0.001$ ) interaction were observed for TOT TUT. Furthermore, the condition by set ( $p = 0.076$ ) approached significance. An increase in TOT TUT was observed over each successive set with set 3 being greater than sets 1 ( $p = 0.002$ ) and 2 ( $p = 0.002$ ) and set 4 being greater than all previous sets when collapsed across condition. Post hoc analysis on the trend observed in the condition by set interaction showed that the TOT TUT to complete each set was lower for all sets in CLU compared with TRD although this difference only approached significance ( $p < 0.10$ ). Post hoc analysis on the condition by rep interaction demonstrated that TOT TUT was greater during TRD at isolated repetitions (rep 1,  $p = 0.052$ ; rep 6,  $p = 0.029$ ; rep 7,  $p = 0.004$ ; rep 8,  $p = 0.004$ ).

Similar to that which was observed in TOT TUT, a main effect of set ( $p < 0.001$ ) and a condition by repetition ( $p < 0.001$ ) interaction were observed for CON TUT. Concentric TUT increased during both conditions progressively from set 1 through set 4, with each set being significantly longer than the previous when collapsed across condition ( $p \leq 0.05$ ). When collapsed over all sets, CON TUT increased significantly from repetitions 1 to 5 in both conditions (Table 3). However, although TRD CON TUT continued to increase through repetition 10, CON TUT during CLU returned to pre-intraset rest values. This resulted in significantly greater CON TUT during TRD in later repetitions.

Eccentric TUT demonstrated a significant main effect for set ( $p = 0.003$ ) and repetition ( $p < 0.001$ ). Only set 4 was



**Figure 3.** Average power output (W) during set 1 (A), set 2 (B), set 3 (C), and set 4 (D) collapsed by training status. \*Significant difference between conditions ( $p \leq 0.05$ ); #Significant difference between conditions ( $p < 0.10$ ).

**TABLE 3.** Concentric time under tension (in seconds) collapsed by training status across sets.\*†

| Repetition | TRD         | CLU          |
|------------|-------------|--------------|
| 1          | 0.97 ± 0.03 | 0.92 ± 0.02‡ |
| 2          | 0.97 ± 0.03 | 0.93 ± 0.02  |
| 3          | 1.02 ± 0.03 | 0.98 ± 0.02  |
| 4          | 1.05 ± 0.04 | 1.03 ± 0.03  |
| 5          | 1.08 ± 0.03 | 1.14 ± 0.03‡ |
| 6          | 1.13 ± 0.04 | 0.96 ± 0.03§ |
| 7          | 1.16 ± 0.04 | 0.97 ± 0.03§ |
| 8          | 1.20 ± 0.04 | 1.04 ± 0.03§ |
| 9          | 1.20 ± 0.05 | 1.11 ± 0.03§ |
| 10         | 1.34 ± 0.06 | 1.25 ± 0.04  |

\*TRD = traditional sets; CLU = cluster sets.

†Values are means ± SE.

‡Significant difference between conditions ( $p < 0.10$ ).

§Significant difference between conditions ( $p \leq 0.05$ ).

**TABLE 4.** Load (in kilograms) by repetition during set 4 collapsed by training status.\*†

| Repetition | TRD         | CLU          |
|------------|-------------|--------------|
| 1          | 82.6 ± 77.3 | 82.6 ± 77.3  |
| 2          | 82.6 ± 77.3 | 82.6 ± 77.3  |
| 3          | 82.6 ± 77.3 | 82.6 ± 77.3  |
| 4          | 82.6 ± 77.3 | 82.6 ± 77.3  |
| 5          | 81.7 ± 76.4 | 82.6 ± 77.3  |
| 6          | 80.3 ± 75.0 | 82.6 ± 77.3‡ |
| 7          | 79.2 ± 72.9 | 82.6 ± 77.3‡ |
| 8          | 77.5 ± 71.2 | 82.6 ± 77.3‡ |
| 9          | 76.1 ± 70.0 | 82.6 ± 77.3‡ |
| 10         | 75.8 ± 70.0 | 82.6 ± 77.3‡ |

\*TRD = traditional sets; CLU = cluster sets.

†Values are means ± SE.

‡Significant difference between conditions ( $p \leq 0.05$ ).

greater than all other sets, with no significant differences reported among other sets. In opposition to that which was observed in TOT TUT and CON TUT, no observable pattern emerged when evaluating post hoc analysis on the main effect of repetition.

#### Load and Total Volume Load

As expected, given their higher 1RM back squat, RT were able to displace a greater load when compared with UT. This resulted in a significantly greater TVL in RT ( $4115.2 \pm 145.2$  kg) compared with UT ( $2462.1 \pm 145.2$  kg;  $p < 0.001$ ). Furthermore, a condition by set by repetition interaction ( $p < 0.001$ ) was observed for load. No significant differences in load were recorded between conditions during sets 1 through 3. However, CLU allowed for maintenance of load throughout set 4, whereas TRD necessitated a reduction in load to permit completion of the exercise bout. Participants were able to displace a significantly greater load during later repetitions of set 4 in CLU (Table 4). This resulted in a significantly greater TVL during CLU ( $3302.4 \pm 102.7$  kg) when compared with TRD ( $3274.8 \pm 102.8$  kg;  $p = 0.001$ ).

#### DISCUSSION

This study demonstrated the beneficial effects of using CLU at intensities associated with hypertrophy (70% 1RM) for RT and UT participants. Specifically, greater average power was produced in the later repetitions of each set and was primarily driven by higher average velocities. Furthermore, despite no previous resistance training history, similar kinetics and kinematics were observed in all participants regardless of current training status or background. Our results are in agreement with previous studies (19,23,29) in which intensities traditionally associated with strength and power were used.

Joy et al. (27) were the first to investigate acute effects of CLU on power output in a hypertrophic scheme. In that

study, RT participants performed the back squat (75% 1RM) using 60 seconds of rest between clusters, CLU produced significantly greater power during later repetitions of each set of CLU when compared with TRD. A similar response was observed in this study as participants experienced a decrease in power output during TRD of  $25.3 \pm 3.4\%$  from repetitions 1 to 10, whereas CLU slowed the decline during later repetitions of a set. It is of interest to note that both UT and RT participants were allowed 30 seconds of rest between clusters of 5 repetitions, which is more consistent with that which has been defined as a typical CLU (17) and thereby demonstrated that short intraset rest periods are sufficient during hypertrophic back squats using CLU configurations.

In contrast to those previous studies (19,23,27) in which greater power output was observed and TVL was equated between conditions, a novel aspect of this study was the reduction in load (13.6 kg) when both RT and UT participants were unable to maintain the prescribed cadence, suggesting failure. The reduction in load required during the TRD condition resulted in significantly greater load during set 4 of CLU at repetitions 6 through 10, as well as significantly greater TVL. It is well established that lower loads require less force and allow for a higher velocity of movement (e.g., force-velocity curve); therefore, it would be assumed that average power during the later repetitions of TRD would be maintained given the lower loads. However, a decline in average power was observed in the TRD condition demonstrated by a decrease in both force and velocity generating capacity of the muscles used.

Maximum isometric force declines progressively with repeated tetani (1,35), and this has been observed in similar studies comparing TRD and CLU. Hardee et al. (23) and Hansen et al. (19) observed a near-linear decrease in force from repetitions 1 to 6 across all sets in TRD. In this study, force output decreased significantly during later repetitions of set 2 and 4, but this was not observed across all sets. However, when collapsed across sets, a similar observation was made, which may be attributable to the exercise selection of the back squat. Participants in previous studies performed the power clean (23) and weighted jump squat (19). When performing the back squat, vertical bar displacement remains constant from repetition to repetition necessitating a certain amount of force to be produced for successful completion of the movement. However, in the in the power clean and weighted jump squat, bar displacement is not consistent across repetitions. Hardee et al. (22) observed a significant decrease in vertical bar displacement from repetitions 1 to 6 in all 3 sets, which decreased the amount of force necessary to displace the bar by that difference.

Sanchez-Medina and Gonzalez-Badillo (34) suggested that a decline in repetition velocity is an indicator of neuromuscular fatigue. It has been proposed that neuromuscular fatigue is caused by a combination of decreased central drive and impaired neuromuscular propagation (30), which can be

partially recovered by rest. This study supports those contentions as movement velocity declined to a greater extent during TRD suggesting greater neuromuscular fatigue. The greater average power observed in this study was attributed to the higher velocities in the CLU condition as when collapsed across sets, average velocity was higher for CLU. This occurred as early as repetition 1. However, upon examination of the trend observed in the condition by set by repetition interaction ( $p = 0.10$ ), it was noted that this difference did not manifest until the third set. This suggests that the additional midset recovery during CLU may have allowed participants to begin later sets at a higher velocity compared with TRD. Furthermore, CLU produced significantly greater power output driven by higher velocities at an increasing number of repetitions from sets 1 to 4. Although the causes of fatigue are likely multifactorial, the results presented herein support the original hypothetical model proposed by Haff et al. (18) in which the authors suggested a brief intraset rest would allow higher velocities to be achieved during later repetitions of a given set, which would thus influence power production.

The slower movement velocities during TRD in turn caused greater CON and TOT TUT when compared with CLU. The greatest effect of movement velocity on TUT was observed in the concentric phase, during which participants were instructed to move “as explosively as possible.” As muscular fatigue increased, CON TUT became greater in both conditions. CON TUT was elevated above repetition-1 values in both CLU and TRD at repetitions 3, 4, and 5. However, after participants received a 30 seconds of break after repetition 5 during CLU, CON TUT returned to initial levels and was significantly shorter than TRD during later repetitions of the set. This divergence caused greater total CON TUT during TRD when compared with CLU. Concentric TUT displayed an inverse relation to mean velocity. As movement velocity declined progressively from set to set in both conditions, CON TUT increased progressively over the same period of time. Reduced load during set 4 of TRD did not have a noticeable effect on CON TUT as CON TUT continued to be greater during TRD even after the load was lowered, suggesting that participants were experiencing fatigue.

Although participants were verbally reminded to perform the concentric phase “as explosively as possible,” no verbal cues were provided for the eccentric portion. This may have caused a high interindividual and intraindividual variability in ECC TUT. The only finding of interest related to ECC TUT was that it decreased significantly in set 4, as muscular fatigue was the greatest in both conditions. Concentric and ECC TUT were lower in both conditions when compared with previously published findings on repetition kinematics and kinetics in a hypertrophic scheme (7). Crewther et al. (7) asked their participants to perform 10 sets of 10 controlled Smith-squat repetitions at 75% of 1RM with 2 minutes of rest between sets. The controlled movement technique elicited ECC TUT averaging  $1.56 \pm 0.11$  seconds and

CON TUT averaging  $1.54 \pm 0.11$  seconds per repetition. In this study, mean CON TUT was  $1.03 \pm 0.03$  seconds during CLU and  $1.11 \pm 0.04$  seconds during TRD, while ECC TUT was  $1.09 \pm 0.05$  seconds during CLU and  $1.11 \pm 0.04$  seconds during TRD. This suggests that attempting to maximize power output in a hypertrophy by moving “as explosively as possible” will result in diminished TUT during both CLU and TRD when compared with a controlled movement technique.

It is commonly accepted that improvements in muscular strength are attributed to a combination of increased muscle cross-sectional area (hypertrophy) and improved neural coordination (6,12,13,33). Mechanical stress is an important factor in both of these adaptations. It has been proposed that the intensity and the duration of muscular tension, or TUT, are key components of maximizing mechanical stress (6). Given the differing reports of long-term adaptation after the performance of CLU in RT and UT participants, these results may provide useful information in determining those differences. Despite the expected lower overall values observed, the pattern and time course of the kinetics and kinematics were similar between RT and UT groups. To the authors’ knowledge, this is the first study to examine the differences in kinetics and kinematics due to training status in TRD and CLU. This study suggests that the duration of muscular tension is augmented in TRD (greater TUT) when compared with CLU. This would suggest greater mechanical stress and thus a greater potential for strength and hypertrophy adaptations after TRD. However, our investigation also indicates that a greater TVL can be achieved with CLU when compared with TRD, thereby maximizing the intensity component of the mechanical stress imposed upon the working muscle. It is reasonable to conclude that CLU configurations have the potential to create similar mechanical stress as TRD, thus facilitating similar strength and hypertrophy adaptations while maximizing power adaptations.

Joy et al. (27) recently demonstrated a significantly different pattern and time course of muscle activity in the vastus lateralis and biceps femoris in RT participants performing the back squat exercise at a similar intensity and condition design to that of this study. The disproportionately greater increases in strength in the absence of any significant increase in muscle size during the early stages of resistance training have been attributed to neurological adaptations (12). It has been suggested that the previously reported (35) greater gains in strength and power were due to neuromuscular adaptations, specifically differences in motor unit recruitment and/or rate coding (27). Whether differences in those underlying mechanisms, which are related to the kinetics and kinematics of CLU, vary between UT and RT is unknown. Additional research is necessary to distinguish the mechanism(s) responsible for the differences in long-term adaptation between RT and UT. However, the similar kinetic and kinematic response regardless of training status

combined with the fact CLU is associated with a lower perception of effort (21) and that a lower perception of effort may increase compliance (2,10) makes CLU a viable training technique in UT populations as well.

## PRACTICAL APPLICATIONS

These findings suggest that velocity drives the greater power observed during back squat exercise using a CLU protocol. Cluster sets allow for achievement of greater movement velocity, power output, and TVL while eliciting similar gains in lean mass to TRD protocols. Thus, CLU can be a valuable tool for strength and conditioning professionals, athletes, and recreational weight lifters. From an overall program design perspective, the use of CLU could allow athletes to emerge from a hypertrophy cycle with similar gains in muscle mass as compared TRD but with greater strength and power gains. Additionally, this might serve to reduce the common decrease in athletic power/velocity associated with hypertrophic training.

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